

Chapter 2 Electrostatics

1. The electric field (E field)

1) Generation of the electrostatic fields

The electric field originates from a. charges, and b. changing magnetic field.

If charges are static and there is no magnetic field $\Rightarrow \vec{E}$ field is static.

2) Measuring the electrostatic fields

In electrostatics, we study the force on a test charge Q due to the distribution of space charges $q_i(\vec{r}_i)$ (i denotes the i 'th charge).

3) Basic principles of electrostatic fields

Everything about electrostatics is based on two principles:

a) The Coulomb's law (which describes the force on Q due to a single charge q):

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{r}$$

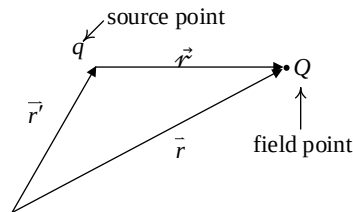


Figure. Relation between field point and source point.

Term-by-term explanation:

- 4π : see Gauss's law below
- ϵ_0 : Permittivity of free space, which is a conversion factor among [coulombs], [meters] and [newtons] (in SI units).

$$\epsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{N \cdot m^2}$$

Note: $1[C] = 1[Ampere] \cdot 1[s]$, $1[e^-] = 1.6 \times 10^{-19}[C]$, $1[C] = 6.2 \times 10^{18}[e^-] \sim 1 \times 10^{-5}[mol]$

Rubbing: $10^{-6}[C]$; Lighting bolt: $15[C] \sim 350[C]$; AA battery: $5000[C]$

- $\frac{1}{r^2}$: like gravity; Maxwell equations $\rightarrow$ Proca's equations: $\frac{1}{r^{2+\delta}}$, a non-zero δ is related to non-zero photon rest mass. For $\delta \sim 10^{-16}$, $m_{photos} \sim 10^{-47}g$.

b) Principle of superposition (when multiple charges exist):

$$\vec{F}_{tot} = \sum_i \vec{F}_i$$

4) The electric field due to multiple charges

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \left(\sum_{i=1}^n \frac{q_i}{r_i^2} \hat{r}_i \right) \cdot Q = Q \cdot \vec{E}, \text{ the electric field is defined as } \vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{r}_i$$

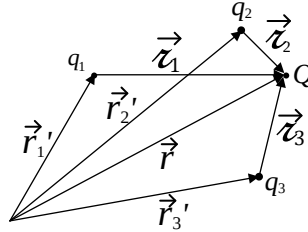


Figure. Superposition principle: electrostatic force due to multiple charges

5) The field due to continuous charge distributions

Extending the expression above, the electric field due to continuous charge distribution is:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{r} dq.$$

Depending on different distribution of charge (line, surface, or volume):

$$dq \rightarrow \underbrace{\lambda dl' \sim \sigma da' \sim \rho dV'}_{\substack{\text{Not a vector.} \\ \text{not "dot" product}}} \quad \left. \begin{array}{l} \text{otherwise, } \hat{r} \cdot d\vec{a} \\ \text{becomes a scalar} \end{array} \right\}$$

For example, for volume charge distribution:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{r^2} \hat{r} dV'$$

where $\rho(\vec{r}')$ denotes the charge density and the integration should include all the charges.

Now, we have the necessary tool to solve all electrostatic problems, bravo!

Example:

A spherical shell carries uniform surface charge distribution σ (unit: [C]/[m²]), the radius of the shell is R , the shell is centered at the origin. P is a point on the z axis. Calculate the electric field at point P .

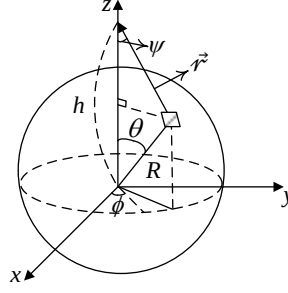


Figure. Illustration of the example problem.

$$dq = \sigma da = \sigma R^2 \sin\theta \, d\theta d\phi, \quad r^2 = R^2 + h^2 - 2Rh\cos\theta,$$

$$\cos\psi = \frac{h - R\cos\theta}{r}, \quad \text{due to symmetry, } \vec{E} = E_z \cdot \hat{z}$$

$$\begin{aligned} E_z &= \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \cdot (\hat{r} \cdot \hat{z}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma R^2 \sin\theta \, d\theta d\phi}{r^2} \cdot \cos\psi \\ &= \frac{1}{4\pi\epsilon_0} \int \frac{\sigma R^2 \sin\theta \, d\theta d\phi \cdot (h - R\cos\theta)}{r^3} = \frac{2\pi\sigma R^2}{4\pi\epsilon_0} \int_0^\pi \frac{(h - R\cos\theta) \sin\theta \, d\theta}{(R^2 + h^2 - 2Rh\cos\theta)^{3/2}} \end{aligned}$$

(set $x = \cos\theta$, $dx = -\sin\theta d\theta$),

$$= \frac{2\pi\sigma R^2}{4\pi\epsilon_0} \int_{-1}^1 \frac{(h - Rx) dx}{(R^2 + h^2 - 2Rh x)^{3/2}} = \frac{1}{4\pi\epsilon_0} \frac{2\pi R^2 \sigma}{h^2} \left(\frac{h - R}{|h - R|} - \frac{(-h - R)}{h + R} \right)$$

$$\text{if } Z > R, \quad E_z = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{h^2}, \text{ where } q = 4\pi R^2 \sigma; \text{ if } h < R, E_z = 0.$$

2. Some properties of the electrostatic field

A practical aspect of the course is to learn the skills to calculate electromagnetic fields. According to the above example, although we can calculate the electrostatic field using Coulomb's law, the computation is too complicated. In fact, the electrostatic field has some favorable properties.

These properties are extremely helpful in field computation.

1) The electrostatic field integrated over a closed surface

a) Point charge inside the closed surface

$$\begin{aligned} \int_s \vec{E} \cdot d\vec{a} &= \frac{1}{4\pi\epsilon_0} \int \frac{q}{r^2} \hat{r} \cdot d\vec{a} \\ &= \frac{q}{4\pi\epsilon_0} \int \frac{1}{r^2} \cdot \hat{r} \cdot \hat{n} \cdot da \\ da' &= da \cdot \cos\theta \\ &= da(\hat{r} \cdot \hat{n}) \end{aligned}$$

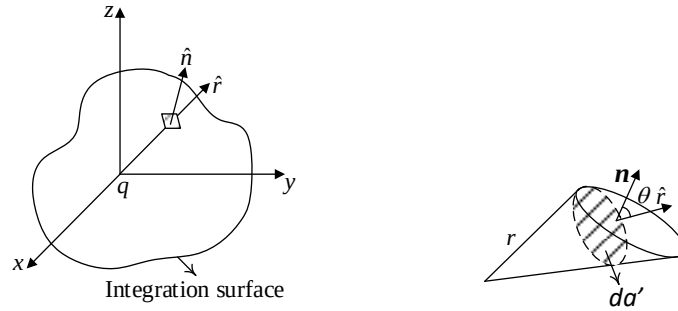


Figure. Illustration of a point charge inside the integration surface.

$$\frac{da'}{r^2} = d\Omega \text{ (solid angle)}$$

$$\rightarrow \int_s \vec{E} \cdot d\vec{a} = \frac{q}{4\pi\epsilon_0} \int d\Omega = \frac{q}{4\pi\epsilon_0} \cdot 4\pi = \frac{q}{\epsilon_0}$$

b) Point charge outside the surface

In this case, we can always find a pair of surface elements whose contributions cancel.

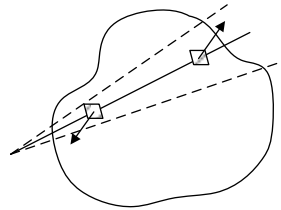


Figure. Illustration of a point charge outside the integration surface

c) Multiple charges:

$$\text{Due to superposition principle, } \oint \vec{E} \cdot d\vec{a} = \frac{\sum_{\text{inside}} q_i}{\epsilon_0}$$

Note: ① this conclusion can be understood as a mathematical representation of the “Conservation of field lines due to the $1/r^2$ law”.

② The E field is still determined by all the charges, including those outside ones.

d) Continuous charge distribution:

$$\boxed{\oint \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enclosed}}}{\epsilon_0}}$$

Note: ① This important conclusion is referred to as the Gauss’s law, which is one of the Maxwell’s equations in the integral form.

② Using the Gauss’s law, it is much easier to calculate $\vec{E}$ if the system has certain symmetry (so the field distribution has certain symmetry). Example: use Gauss’s law to solve the problem in the

previous example.

2) Divergence of the electrostatic field

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{r}}{r^2} \rho(\vec{r}') dV' \quad \leftarrow \text{Coulomb's law}$$

According to the above Coulomb's law, we can directly compute the divergence:

a) The first approach:

$$\nabla \cdot \vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \nabla \cdot \left(\frac{\hat{r}}{r^2} \right) \rho(\vec{r}') dV'$$

Note:

① $(\nabla \cdot)$ acts on $\vec{r}$ but not $\vec{r}'$, so $\rho(\vec{r}')$ is treated as a constant.

$$\textcircled{2} \quad r = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$$

$$\hat{r} = [(x - x')\hat{x} + (y - y')\hat{y} + (z - z')\hat{z}]/r$$

$$\textcircled{3} \quad \nabla \cdot \left(\frac{\hat{r}}{r^2} \right) = 4\pi\delta^3(\vec{r})$$

$$\text{As a result, } \nabla \cdot \vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \cdot 4\pi \int \delta^3(\vec{r}) \rho(\vec{r}') dV'$$

$$\vec{r} = \vec{r} - \vec{r}', \text{ so } \vec{r}' = \vec{r} - \vec{r}$$

Note: the above equation involves the sifting property of the delta function. As a review, let's consider the one-dimensional problem to evaluate

$$\int_{-\infty}^{\infty} \delta(T - t) p(t) dt$$

Let $x = T - t$, $t = T - x$, $dt = -dx$, $\delta(T - t) = \delta(x)$

$$\int_{-\infty}^{\infty} \delta(T - t) p(t) dt = \int_{\infty}^{-\infty} \delta(x) P(T - x) (-dx) = \int_{-\infty}^{\infty} \delta(x) P(T - x) dx = P(T)$$

$$\text{OR: } \int_{-\infty}^{\infty} \delta(T - t) p(t) dt \xrightarrow{\delta(x) = \delta(-x)} \int_{-\infty}^{\infty} \delta(t - T) P(t) dt = \int_{-\infty}^{\infty} \delta(x) P(x + T) dx = P(T).$$

From the above analysis, $\vec{E}(\vec{r}) = \frac{1}{\epsilon_0} \int \delta^3(\vec{r}) \rho(\vec{r}') dV'$ can be simplified using the sifting property of the delta function:

$$\boxed{\nabla \cdot \vec{E}(\vec{r}) = \frac{1}{\epsilon_0} \rho(\vec{r})}$$

This is the Gauss's law in differential form, which is also one of the Maxwell's equations.

b) The second approach:

$$\text{From Gauss's law: } \oint \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} Q_{enc}$$

$$\oint \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} \int \rho dV$$

Using the divergence theorem,

$$\int (\nabla \cdot \vec{E}) dV = \oint \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} \int \rho dV$$

$$\int_V \left[\nabla \cdot \vec{E} - \frac{\rho}{\epsilon_0} \right] dV = 0 \quad \text{for ANY volume } V$$

$$\Rightarrow \boxed{\nabla \cdot \vec{E} = \rho/\epsilon_0}$$

3) The electrostatic field integrated along a closed loop

Let's first consider the electric field due to a point charge:

$$\vec{E}(\vec{r}) = \frac{q}{4\pi\epsilon_0} \frac{\hat{r}}{r^2} \quad (\vec{r} = \vec{r} - \vec{r}')$$

Since the position of the integration loop and the point charge is relative, we can shift the coordinate system to place the point charge at the origin.

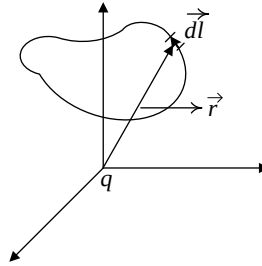


Figure. Illustration of the integration loop.

In spherical coordinates, $d\vec{l} = dr \hat{r} + r d\theta \hat{\theta} + r \sin\theta d\phi \hat{\phi}$

$$\vec{E} \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0} \frac{\hat{r}}{r^2} \cdot (dr \hat{r} + r d\theta \hat{\theta} + r \sin\theta d\phi \hat{\phi})$$

$$= \frac{q}{4\pi\epsilon_0} \frac{dr}{r^2}$$

$$\Rightarrow \oint \vec{E} \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0} \int_{r_o}^{r_o} \frac{dr}{r^2} = 0$$

$\Rightarrow \int_a^b \vec{E} \cdot d\vec{l}$ is independent of the path taken, which means the work done by the electrostatic force is independent of the moving path of the charge, and is only determined by the end points. In other words, the electrostatic field is conservative.

To prove the above conclusion, suppose $\oint \vec{E} \cdot d\vec{l} = 0$, but $\int_a^b \vec{E} \cdot d\vec{l} = X$ when integrated along

path A, while $\int_a^b \vec{E} \cdot d\vec{l} = Y$ when integrated along path B, $X \neq Y$.

$$\therefore \int_{\substack{\vec{a} \\ \vec{b} \\ \text{path } B}} \vec{E} \cdot d\vec{l} = -Y \quad (d\vec{l} \text{ is inversed everywhere})$$

$$\therefore \int_{\text{along } A}^{\vec{a} \rightarrow \vec{b}} \vec{E} \cdot d\vec{l} + \int_{\text{along } B}^{\vec{b} \rightarrow \vec{a}} \vec{E} \cdot d\vec{l} = X - Y \neq 0.$$

However, $\int_{\text{along } A}^{\vec{a} \rightarrow \vec{b}} \vec{E} \cdot d\vec{l} + \int_{\text{along } B}^{\vec{b} \rightarrow \vec{a}} \vec{E} \cdot d\vec{l} = \oint \vec{E} \cdot d\vec{l} = 0$, which is in contradiction to the equation above. As a result, $\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l}$ cannot be dependent on the path taken from $\vec{a}$ to $\vec{b}$.

Suppose $\vec{E}$ is the electric field generated by an arbitrary charge distribution, due to the superposition principle, $\oint \vec{E} \cdot d\vec{l} = 0$ always applies.

4) The curl of electrostatic field:

a) The first approach:

Consider the field generated by a single charge, then extend the conclusion to arbitrary charge distribution applying the superposition principle. For a single charge, its electric field is

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

Compute its curl in spherical coordinates:

$$\begin{aligned} \nabla \times \vec{E} &= \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta E_\phi) - \frac{\partial E_\theta}{\partial \phi} \right] \hat{r} \\ &+ \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial E_r}{\partial \theta} - \frac{\partial}{\partial r} (r E_\phi) \right] \hat{\theta} \\ &+ \frac{1}{r} \left[\frac{\partial}{\partial r} (r E_\theta) - \frac{\partial E_r}{\partial \theta} \right] \hat{\phi} \end{aligned}$$

$$\therefore E_\phi = E_\theta = 0, \quad E_r = E_r(r) \text{ (not a function of } \theta, \phi)$$

$$\therefore \boxed{\nabla \times \vec{E} = 0} \text{ for all possible charge distributions.}$$

b) The second approach:

$$\oint_{\text{closed loop}} \vec{E} \cdot d\vec{l} \xrightarrow{\text{Stokes Theorem}} \int_{\text{Any surface attached to the loop}} (\nabla \times \vec{E}) \cdot d\vec{a} = 0$$

Now, we are left with the following question:

$$\int (\nabla \times \vec{E}) \cdot d\vec{a} = 0 \quad \begin{matrix} \Leftarrow \sqrt{} \\ \Rightarrow ? \end{matrix} \quad \nabla \times \vec{E} \equiv 0$$

Any surface

To prove “ $\Rightarrow$ ” holds, let the loop to be very small, and the integration surface to be an infinitesimal area surrounding $\vec{r}$ ($\vec{r}$ denotes the location where $\nabla \times \vec{E}$ is evaluated).

$$\int (\nabla \times \vec{E}) \cdot d\vec{a} \xrightarrow{\text{when area is very small}} (\nabla \times \vec{E}(\vec{r})) \cdot \int d\vec{a} = (\nabla \times \vec{E}) \cdot \vec{A} \text{ where } \vec{A} \text{ is the small area.}$$

Because $(\nabla \times \vec{E}) \cdot \vec{A} \equiv 0$ while $\vec{A}$ is not zero, the only possibility is that $\nabla \times \vec{E} \equiv 0$.

So we have proven

$$\boxed{\nabla \times \vec{E} = 0}$$

Note: since we have demonstrated that $\oint \vec{E} \cdot d\vec{l} \equiv 0$, one may naively reach the conclusion that $\vec{E} \equiv 0$ using a similar approach as above. However, this is incorrect: let $\vec{l}$ be a very small loop surrounding $\vec{r}$, similarly, $\oint \vec{E} \cdot d\vec{l} \approx \vec{E} \cdot \oint d\vec{l}$. However, $\oint d\vec{l} \equiv 0$, so from

$$\oint \vec{E} \cdot d\vec{l} \approx \vec{E} \cdot \oint d\vec{l} = \vec{E} \cdot 0 = 0 \quad \text{we cannot conclude } \vec{E} = 0!$$

Actually, $\vec{E}$ is apparently not 0 while $\oint \vec{E} \cdot d\vec{l} = 0$ always holds.

3. Electric (scalar) potential

1) Definition:

We can readily prove if $\vec{E} = -\nabla V \Rightarrow \nabla \times \vec{E} = 0$ (using the identity we've proven in the first chapter). Now that we know $\nabla \times \vec{E} = 0$, can we confidently say that $\vec{E}$ is the gradient of some scalar function, such that $\vec{E} = -\nabla V$? The answer is yes.

$$\text{Proof: } \nabla \times \vec{E} = 0 \quad \Rightarrow \quad \int_{\text{Any surface}} (\nabla \times \vec{E}) \cdot d\vec{a} = \oint_{\text{Any loop}} \vec{E} \cdot d\vec{l} = 0$$

Define $V(\vec{r}) = - \int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{l} \quad \leftarrow \text{Any path}$

$$\underbrace{V(\vec{b}) - V(\vec{a})}_{\text{Any path}} = - \int_{\infty}^{\vec{b}} \vec{E} \cdot d\vec{l} + \int_{\infty}^{\vec{a}} \vec{E} \cdot d\vec{l} = \int_{\vec{b}}^{\infty} + \int_{\infty}^{\vec{a}} = \int_{\vec{b}}^{\vec{a}} \vec{E} \cdot d\vec{l}$$

$$= - \int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l}$$

Any path

$$\int_{\vec{a}}^{\vec{b}} (\nabla V) \cdot d\vec{l} \equiv \int_{\vec{a}}^{\vec{b}} (-\vec{E}) \cdot d\vec{l} \quad \Rightarrow \quad \vec{E} = -\nabla V$$

Note:

① Physical meaning of $V(\vec{r})$: work done by the electric field to move a unit charge from $\vec{r}$ to infinity. V is termed the scalar potential of the electric field.

② Benefit of introducing the scalar potential: simplify the complicated computation of a vector function $\vec{E}$ to the much simpler problem of calculating a scalar function V . Once V is determined, $\vec{E}$ can be unambiguously fixed.

③ Potential obeys the superposition principle.

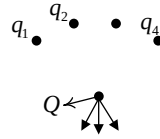


Figure. The force exerted on the test charge Q due to several source charges.

$$\begin{aligned}\vec{F}_{tot} &= Q\vec{E}_1 + Q\vec{E}_2 + Q\vec{E}_3 + Q\vec{E}_4 \\ \Rightarrow \vec{E}_{tot} &= \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4 \\ V &= - \int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{l} = - \int_{\infty}^{\vec{r}} (\vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4) \cdot d\vec{l} \\ &= V_1 + V_2 + V_3 + V_4\end{aligned}$$

2) The electric potential as a function of charge distribution:

For the field of a single charge,

$$\begin{aligned}\vec{E} &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \Rightarrow \text{Move to origin} & E &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \\ V(\vec{a}) &= - \frac{q}{4\pi\epsilon_0} \int_{\infty}^{\vec{a}} \frac{\hat{r}}{r^2} \cdot \underbrace{\hat{r} dr}_{\text{integrate radially}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \Big|_{\infty}^a = \frac{1}{4\pi\epsilon_0} \frac{q}{\underset{\text{distance to } q}{a}} \\ \therefore V(\vec{r}) &= \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad r: \text{distance from the field point to the charge}\end{aligned}$$

For the general case of volume charge distribution, we can apply the superposition principle:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{r} dV'$$

For the more general case of line, surface, and volume charge distribution,

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$$

3) Poisson's equation:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \quad \vec{E} = -\nabla V \Rightarrow -\nabla^2 V = \frac{\rho}{\epsilon_0}$$

$$\boxed{\nabla^2 V = -\rho/\epsilon_0}$$

$\Rightarrow$

$$\boxed{\nabla^2 V = 0}$$

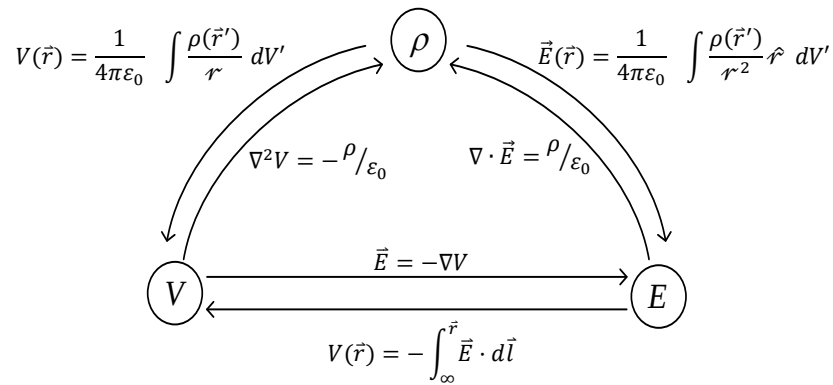
Poisson's equation for $\rho = 0$ region Laplace's equation

4) Summery:

① Units: $q: [Coulomb], F[Newton], [E] = \frac{[F]}{[q]} = \frac{[Newton]}{[Coulomb]}$

$$V = - \int_{\infty}^{\vec{a}} \vec{E} \cdot d\vec{l} = \frac{[Newter] \cdot [Meter]}{[Coulomb]} = \frac{[Joule]}{[Coulomb]} = [Volt]$$

② Relation among E , V , and ρ :



③ Are we ready to proceed to calculating $\vec{E}$ in real situations? Not yet. In most cases, ρ is unknown. We must learn the properties of different types of materials, and the behavior of the electric fields at the boundary.

4. Conductors

In the context of classical electrodynamics, conductors are treated as materials having an infinite supply of free charges. Here, free charges refer to electrons or ions that are free to move when subject to an electric field. For a perfect conductor in a stable state (i.e., supporting an electrostatic field), the following properties apply:

- ① $\vec{E} = 0$ everywhere inside the conductor. Otherwise, free charge will move, which contradicts the assumption of equilibrium state.
- ② $\rho = 0$ everywhere inside the conductor. Since $\vec{E} = 0$ everywhere, $\rho = \epsilon_0(\nabla \cdot \vec{E}) = 0$. Note: this only means there is no net charge, in fact, there are equal amount of positive and negative charges inside.
- ③ All net charge resides on the surface (if you feel confused, no worries. You'll understand this after learning the boundary conditions).
- ④ The whole conductor must be an equipotential. This is because $\nabla V = -\vec{E} = 0 \Rightarrow V \equiv \text{Constant}$
- ⑤ The E field outside the conductor must be perpendicular to the surface, otherwise charges will flow along the surface, which is against the assumption that the conductor is in equilibrium state.